Natural Complexity. An Introduction to Structural Design with Tree Forks: Innovative Techniques of Representation in Architectural Design

Chapter · February 2019

DOI: 10.1007/978-3-030-03676-8_29

2 authors:

Lukas Allner
University of Applied Arts Vienna

Daniela Kroehnert
University of Applied Arts Vienna

Natural Complexity. An introduction to structural design with tree forks[1]

Lukas Allner* and Daniela Kroehnert*
*University of Applied Arts Vienna, lukas.allner@uni-ak.ac.at, daniela.kroehnert@uni-ak.ac.at

Abstract:

Amongst construction materials wood offers a unique potential for deriving complex spatial geometries from its grown fibrous structure. The inherent material properties such as grain direction, can be used as generative parameters for innovative timber structures. With the possibilities of 3D- scanning and Computer Tomography, the anatomy of wood can be exploited as a design driver for spatial structures. This chapter introduces a design concept of utilizing natural forked branches as components in structural frameworks.

Keywords: "3D Scan, Naturally grown form, Complex Wood Structures, Spatial frameworks, discrete element aggregation"

1. Introduction

Trees naturally form intricate configurations of roots, branches, leaves etc. that correspond to multiple forces and conditions. As a result, wood is a dynamic inhomogeneous material with varying densities, directional grain and a complex geometry.
Usually, trunks are cut into rectilinear elements and the more irregular parts of the tree are chipped and processed further into timber products. The resulting standardized components have the highest possible homogeneous quality, in order to assemble structures according to the logic of repetition of identical parts, mainly in the form of grids and other regular organizational principles.
However "the resulting timber-products are showing substantial deficiencies in comparison to the undamaged biological system 'Tree'" (Müller, Gindl-Altmutter, Konnerth, Kaserer, Keckes, 2014). This practice of homogenizations is nevertheless today's standard, since the components can be optimally manufactured and controlled by industrial machines and because it corresponds to a design method in which a spatial scheme is first developed with a virtual (digital) abstract geometry and then adapted to a building material.

Advanced technologies and methods are generally driven by digital abstraction. The physical translation commonly aims to be as close as possible to the mathematical pure geometry as an ultimately virtual "ideal". Complex geometries and organic freeform shapes eg. beams of the Centre Pompidou Metz (Ban, 2010) are often realized as elements that are cnc cut from solid blocks with a wasteful material utilization.

We believe wood offers a potential for an aesthetic of complexity by its heterogeneous material properties and its grown form, but also for novel structural models.
With the possibilities of 3D-scanning and Computer Tomography, the anatomy of wood can be activated and exploited as a design driver for spatial structures closely related to the material logic. In fact, we intend to develop conceptually organic structures that are directly derived from the natural properties of the grown material and which, despite forming artificial compositions, are nevertheless closer to reality than to virtuality.

This article introduces 'Branch Formations' an investigation in the context of the research project "Conceptual Joining" at the University of Applied Arts in Vienna, funded by the Austrian Science Fund (FWF). The aim of this project is to explore the relationship between material and space and more specifically with an intention to translate the grown complexity of wood into architectural experience (Fig. 1).
Branch Formations investigates the use of grown forked branches as structural components in spatial frameworks to take advantage of these naturally optimized nodes.

A reference is the "Wood Chip Barn" - project of the Architectural Association from 2016 (Self, Vercruysse, Students, 2016) where natural crotches of beech were used to compose the primary structure of a roof.
In "Branch Formations" the potentials of grown form are exploited independent from a specific structural function. With this agenda this project navigates in-between the uniqueness of a spatial configuration affected by the singular characteristic of each branch element and the similarities within types of trees.
The use of forked branches as naturally optimized structural nodes allows us to design filigree structures with minimal cross-sections. This advantage of maximum structural performance at a minimum weight has already been employed for tools in pre-industrial times (Spindler, 1995). Some trees were even cultivated with a designed geometry (Agote Aiziprua, 2009).

---

[1] This Article is based on a paper presented at the IASS Conference 2018 in Boston: L. Allner, D. Kroehnert (2018) Conceptual Joining: Branch Formations, Proceedings of IASS 2018: Creativity in Structural Design

Today with computer aided technologies we have the capability to look at wood differently, to liberate its nature and invent solutions beyond industrial standards.

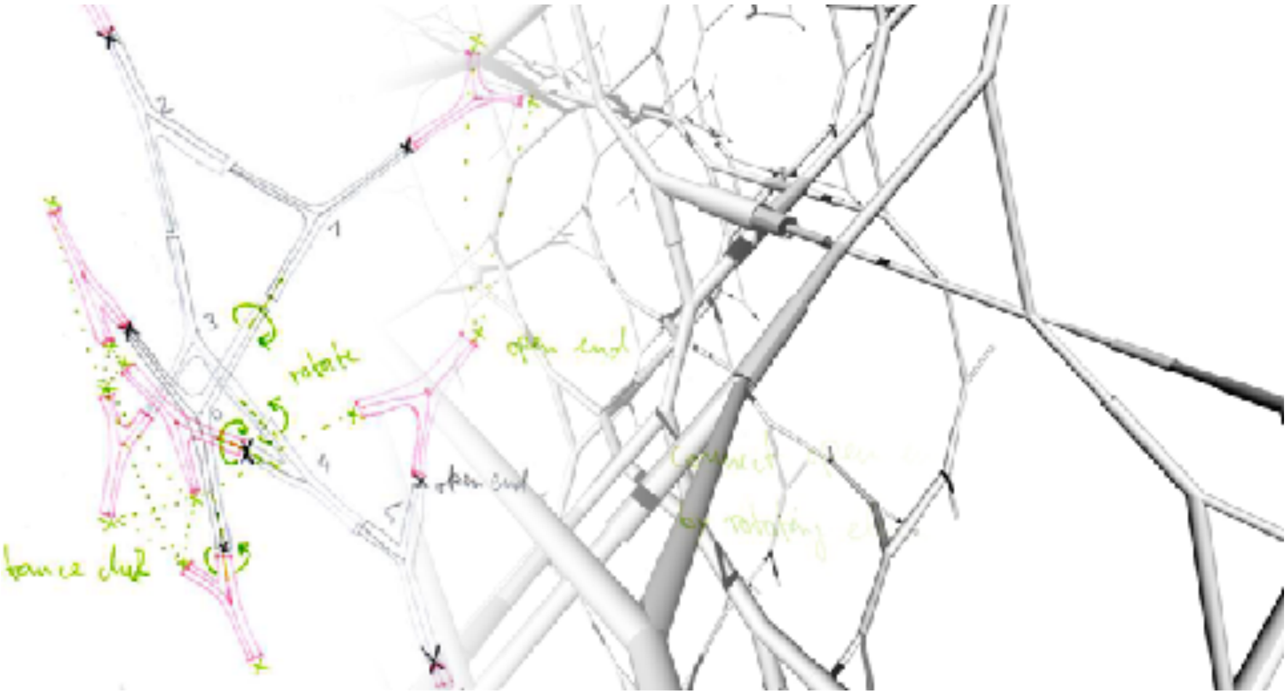


Fig. 1: concept

2. Aims

The research presented in this chapter attempts to understand the complex and irregular parts of trees as a potential. The wood industry considers these as disadvantageous, because branch nodes disrupt the desired homogeneity and structural performance of linear timber profiles. With an alternative concept of following the inherent logic of wood elements and utilizing them accordingly we will be able to activate a hidden potential and eventually transform the presumed disadvantage into a benefit. This would increase the value of non-standard tree parts — they no longer serve only as a source material for chip- and fiber-boards or fuel — and activate their potential for construction.
Our research represents a quest to develop rulesets for assembly as well as design methods for the use of these parts in structural architectural schemes.
The present text describes an ongoing research and is partially a report of undertaken surveys but it also includes speculations on possible outcomes and future investigations.

The dialogue and tension between control (design intention) and coincidence (natural singularity) is to be negotiated constantly.

3. Digital Nature

3D Scanning Technology and computational tools allow us to cope with the complexity of naturally grown unique elements such as branch nodes and consider a “non-standard approach” (Schindler, Tamke, Tabatabai, Bereuter, 2013) to structural design.

3.1. *Data Capturing*

As an approach to designing structures using natural tree parts we choose a method of simplification in a digital model. The geometrical complexity of the individual pieces is reduced to a minimum maintaining the basic typological principles in an axis model which serves as reference for parametrisation. The basis of this method of working is a digitized catalogue of 3D scanned branch nodes.
Tools such as Microsoft ™ Kinect, Artec Eva and photogrammetric software capture geometry as textured 3D envelopes. For the studies presented here this method is providing the basic data in the form of 3D mesh geometry and a sufficient amount of information to approximate the digitized elements’ structural performance. Computer Tomographic Technology (eg. Microtec CT Log) additionally registers internal structures, offering more advanced data sets for future investigations.

Evaluating branch forms in various tree species suggest the use of hard wood branch nodes as architectural and structural components. Deciduous trees have branches of larger diameters in relation to the trunk than those of conifers and they form crowns which are usually not yet considered for construction. Crown parts

represent a considerable proportion of the overall tree and because of the slower growth rate of hard wood, concepts for activating this unused potential are most relevant. Additionally hard wood will become increasingly relevant to the wood industry in middle Europe due to climate change, that requires the consideration of wood species other than those that have prevailed so far.
The most commonly used spruce is more sensitive to increasingly warm weather with longer dry periods than other trees and stocks are expected to shrink in the coming decades. Various types of deciduous trees on the other hand are more robust and thus represent a valid and perhaps necessary alternative (Schüler, Grabner, Karanitsch-Ackerl, Fluch, Jandl, Geburek, Konrad, 2013). Various studies suggest that in the future, trees of that category such as beech will become ever more important for the timber industry in Europe (Hanewinkel, 2010).
Utilizing complex tree parts for structural design additionally offers the potential to consider a greater variety of wood types; species that offer good structural material properties but have been so far disregarded due to their complex shape, including trees with shorter trunks or shrubs.

For the present studies we are working with branches of Hornbeam (carpinus betulus) which is characterised by a high rigidity and a tendency to extensive branching. Therefore it was the favoured tree type used already by the Romans to grow natural ramparts (parts of the Limes). In middle Europe it was used in combination with beech and oak until the late 1800 for growing “Landwehr” (Lüster, 1913). The wood of this tree has also been used for tool grips and mechanical parts in preindustrial times but doesn’t play a relevant role in contemporary industry mainly due to its complex shape and a relatively small diameter of its trunk (Grabner, 2017).

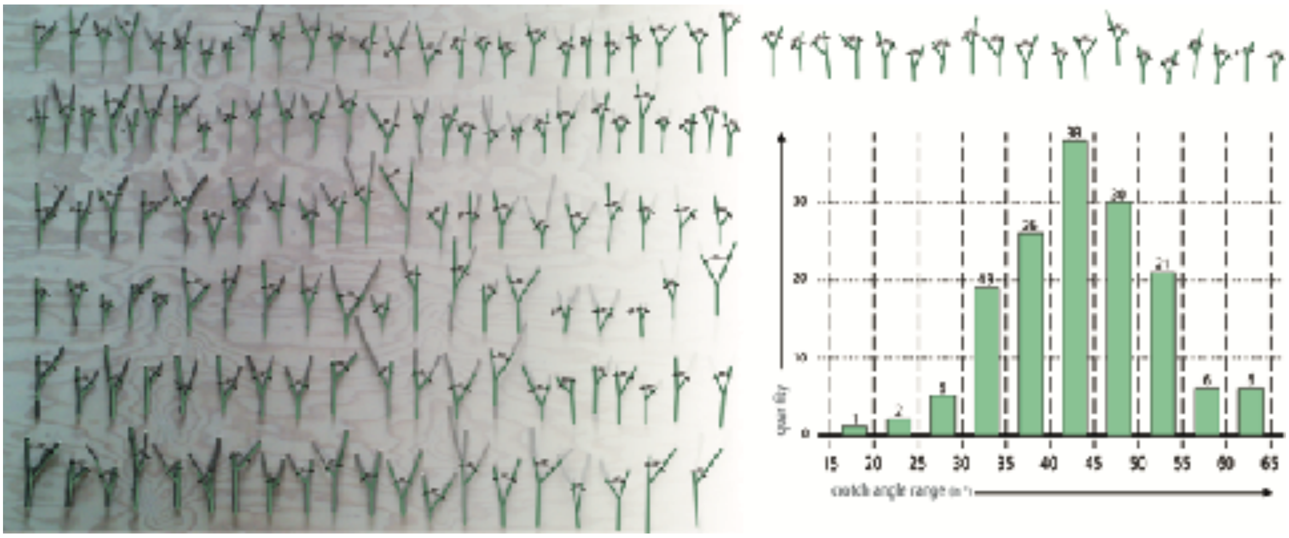


Fig. 2: crotch angle range of hornbeam

In empirical analysis we observed in a sample set of 154 Y-branches an average crotch angle of 42.7°, ranging from a minimum of 16.8° to a maximum of 62.2°, of which 87% are within a range of 30°- 55° (Fig. 2). The set contains elements that are unequal but very similar. The same range of angles occurs across all generations and scales of branches within the sample tree crown which allows for tests in models of small scale to approximate geometric properties of full scale configurations.

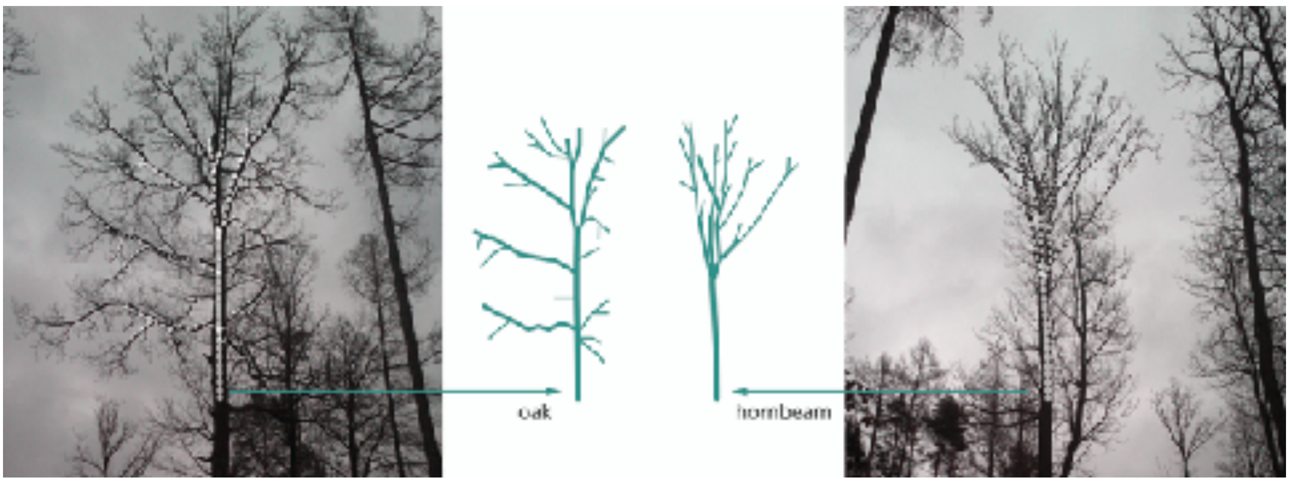


Fig. 3: comparison: oak - hornbeam

In other tree types such as Oak the average crotch angle as well as minima and maxima are different, resulting in a diverging crown architecture (Fig 3). There is very little research or scientific data on typical crotch angles and statistical distribution for various tree species. We assume, that angles are varying only in a limited range within one species under similar growth conditions. Across species on the other hand there is a large variety of angles which will affect the geometrical and structural configurations formed by the natural branch nodes.

3.2. *Abstraction Principles and Control Geometry*

A given set of forked branches is 3D scanned and the digitized data simplified into approximated axis models. Each branch can be represented by 4 points and 3 lines. The node-element represented in modules with three line segments is used as a 'Y' - typology (Fig. 4).

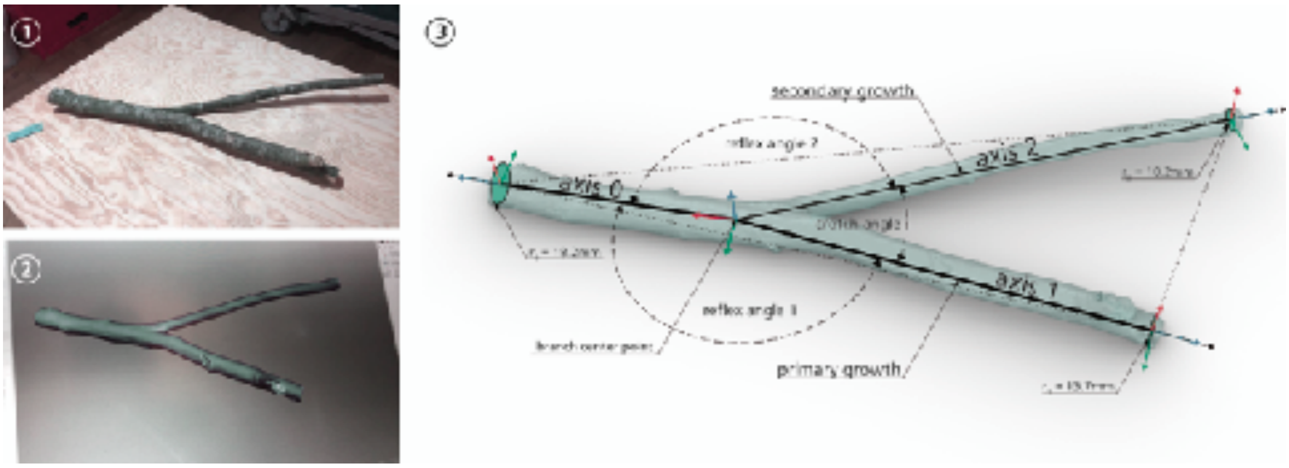


Fig. 4: from 3D scan to axis model (version a)

The axis models are used as components in a parametric model. They also serve as reference geometry to eventually reposition the original scan models into the assembled configuration and thus allow for designing specific joints and modification of the physical parts further along in the process of our research.

There are different possible approximations/translations of branch nodes into Y-models (Fig. 5):

a) Nodes with variable branch lengths. Each branch is approximated as one axis line.

b) Nodes with branches in varying length. End segments are approximated as an axis lines, interpolated within branch node.

c) An inscribed circle around the center point trims branches to same lengths. Branches are represented as approximated axis lines.

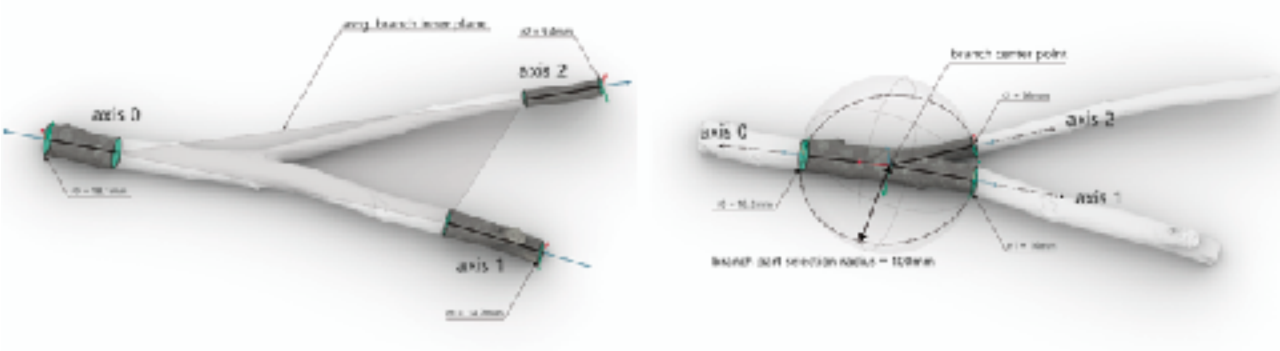


Fig. 5: additional abstraction principles (version b + c)

The axes in version c) are nearly planar, whereas version a) and b) mostly result in 3-dimensional models.
All of these approximations result in similar simplified digital typology and are conceptually interchangeable in the digital assembly model. But the difference in their relationship to the respective physical origin offers different options of a physical translation of the digital assembly in the subsequent steps.

4. Aggregation Principles

To stay close to the principle of the tree with its layout of continuous (bifurcating) fibers we set up an assembly concept in which branch nodes are joined in extension of the wood grain to form a potentially endless continuum (Fig. 6). We aim to develop spatial frameworks for architectural purpose made from wooden parts exclusively, but without being yet tied to a specific use. To allow for a maximum flexibility and range of possible applications we conceptualize configurations of Y-elements with their open ends joined to form closed cells to provide general structural integrity and bracing. This offers possibilities for multidirectional orientation and load bearing capacity by maintaining the objective of fiber and grain continuity. The emerging configurations assemble the Y-elements into bifurcating space loops.
In a first series of studies we examine possible geometrical formations and methods of control and generative logic.

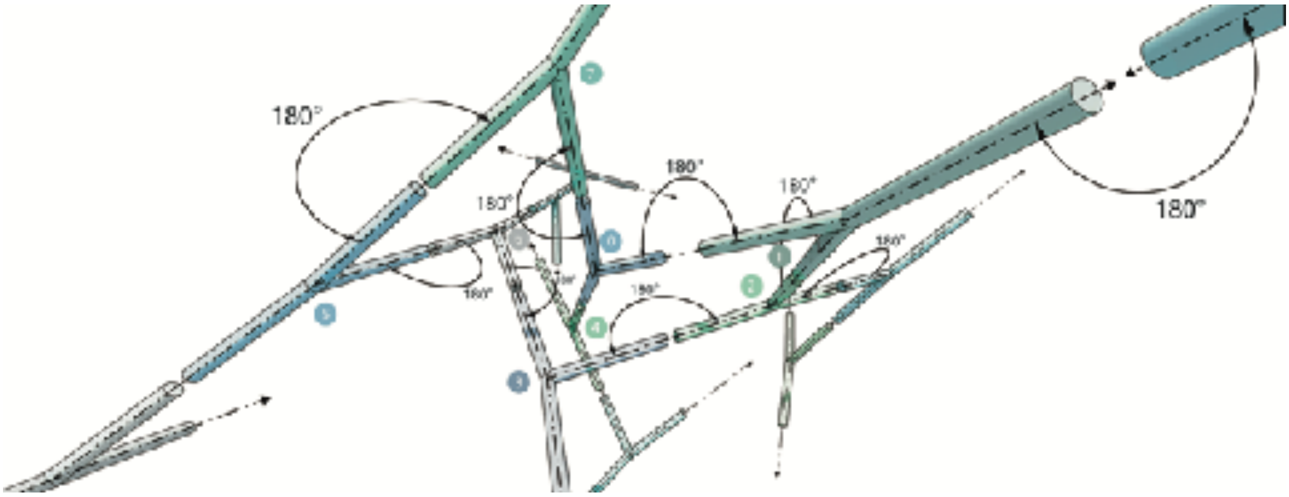


Fig. 6: branch-forks connection principle

4.1. *Discrete Element Aggregation*

Based on findings from intuitive physical and digital models (Fig. 7) we decided to distinguish three basic principles for aggregations: Repetitive patterns (a), Iterative aggregation / self organisation (b) and Population of a designed structure

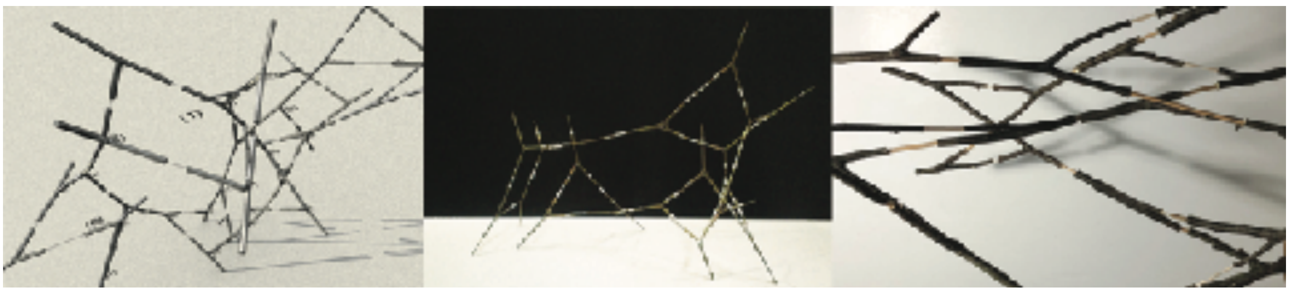
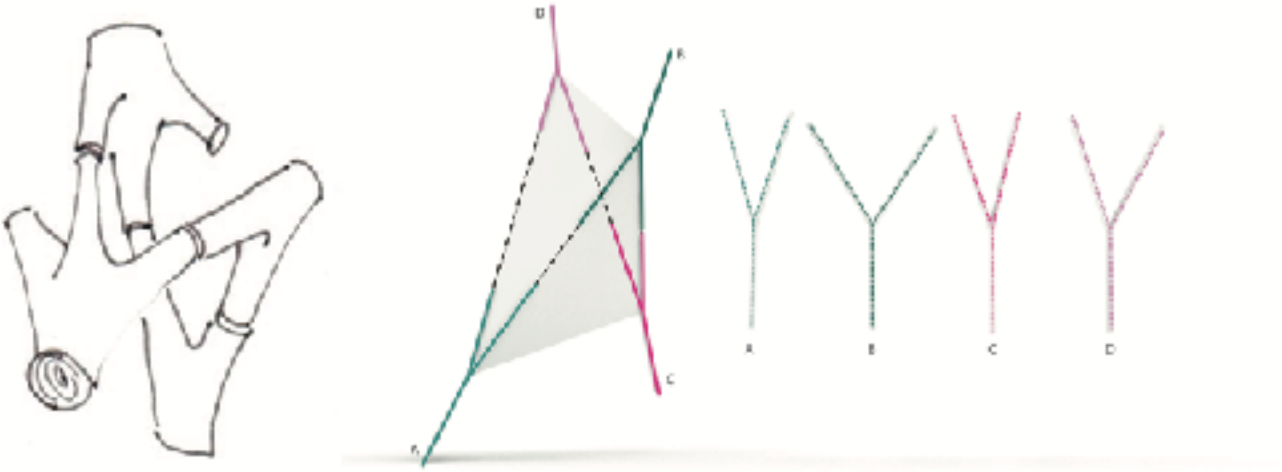


Fig. 7: intuitive studies

a) Repetitive patterns:

Offering the highest level of control and predictability this principle was our starting point and is mainly in focus of our investigation up until this point in the project.

A computational process is set up for aggregating repetitive and regular structures based on mathematical-geometrical principles. Depending on the defined aggregation rule set 2 and 3-dimensional cellular patterns can be formed using the branches as edges. From all possible cell geometries a Tetrahedron is the most compact 3 dimensional form.
The tetrahedron is the platonic solid with the least amount of faces. Thus it also represents the most compact spatial cell with the highest stiffness formed by y-shaped branch nodes (Fig. 8). As a result



aggregations of tetrahedrons (Fig. 9) are the most stable regular spatial frameworks for the chosen connection concept of fiber continuity.

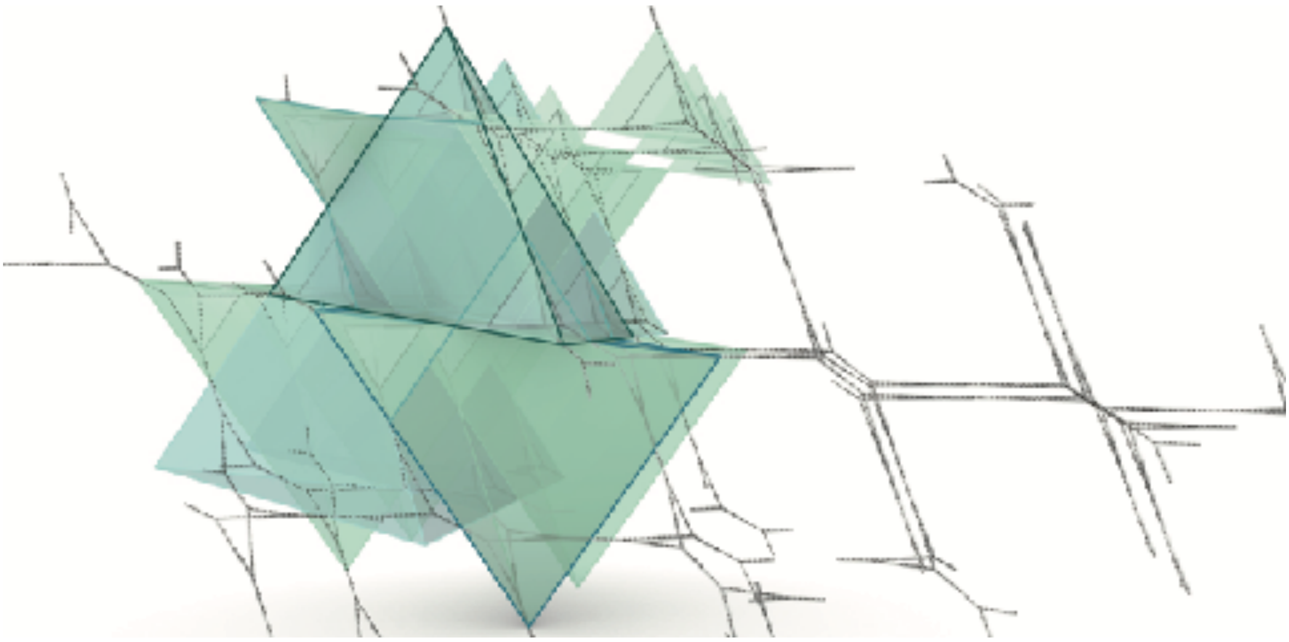


Fig. 8: tetrahedron module from Y-elements

.

Fig. 9: tetrahedron framework

Rotation and length relating to the branch axis represent sufficient levels of freedom necessary to combine elements of any angle into a Tetrahedron cell when considering a set with crotch angles ranging from 17°-62°.

Forming platonic solids or other 3 dimensional cells with more faces and edges would allow for a larger flexibility of possible combinations (Fig. 10). But the increased flexibility also results in forms of a less stiffness.
In a different set with a larger angle range combinations of extremely differently angled elements would eventually fail to form a continuous framework.

With the Wasp Plugin (Rossi, 2017) for the 3D modeling environment Rhinoceros/Grasshopper (McNeel, Rutten), regular frameworks of tetrahedrons can be achieved by duplicating and aggregating one Y-branch element applying the respective ruleset. Current design practice in working with discrete element aggregation is mainly working with a limited amount of different digital entities within the aggregation processes. In this project however we are confronted with a large amount of unequal but similar objects. To cope with this challenge we introduce an operation in which each instance of the identical duplicates is randomly replaced by one (simplified) representation of a “real” specimen from the digitized catalogue, each with a different crotch/reflex angle. The resulting discontinuous pattern is modified with the employment of a relaxation algorithm using Kangaroo for Grasshopper (Piker, 2017). The algorithmic definition is programmed with constrained angles within each Y-element, variable length/distance between modules and a goal set to align adjacent branch axis. The simulation results in a slightly modified irregular pattern of identical typology but with adjusted edge-lengths and an angle variance stemming from the scanned catalogue (Fig. 11).

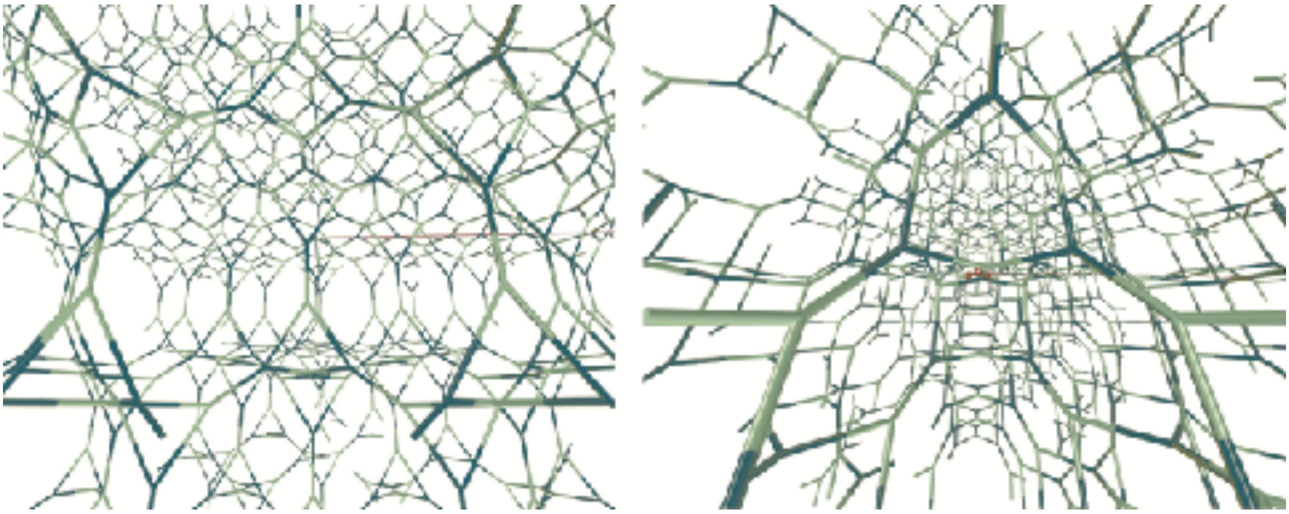


Fig.10: tetrahedron & octahedron framework

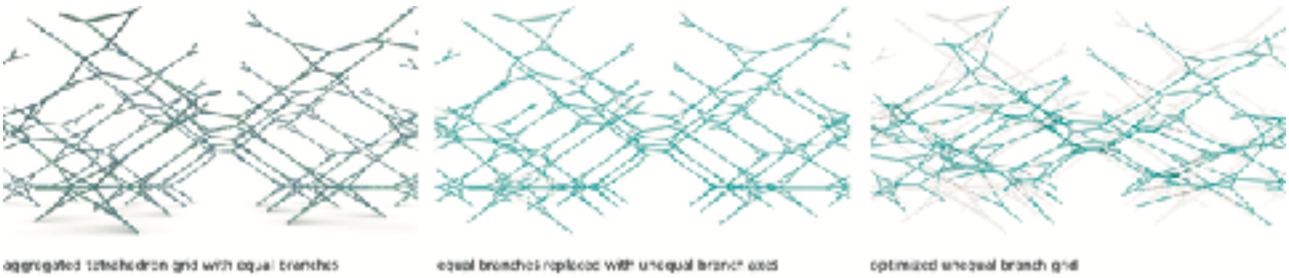


Fig.11: tetrahedron framework: regular - mismatching - optimized

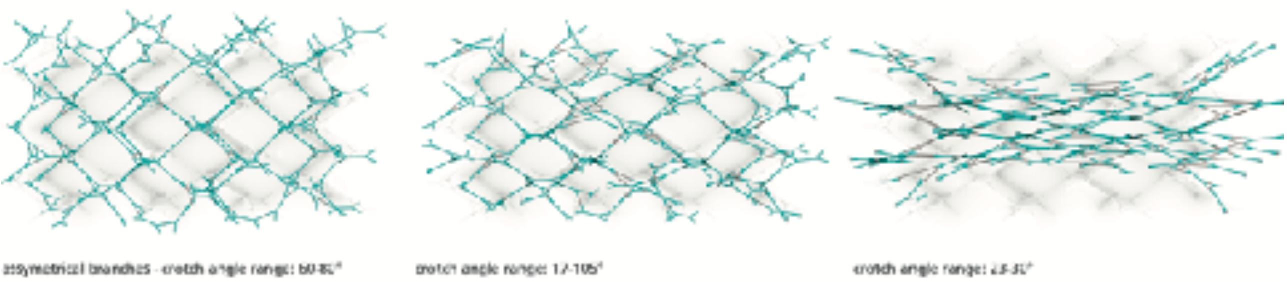


Fig.12: tetrahedron frameworks from different sets of angles

In a first step a random positioning of elements is used as means of handling formations composed of sets of unequal elements. It is one instance of a design setup in a process of negotiating levels of control and design intention on one hand and coincidence introduced by the variance of natural elements on the other hand. In subsequent steps we aim at introducing means of control according to local and global design goals and conditions.

Sets with varying ranges of angles representing branches from different tree species would result in very different appearances (Fig. 12) and structural behavior (when disregarding material specificity). This shows a direct relationship of specific typical growth form related to a species and the resulting spatial formation. In these structures the distinct material property of wood represented in geometrical form becomes spatial and thus architectural experience. The palette of perception of wood structures is expanded beyond the sensing of visual and haptical quality of architectural surfaces. This spatial material specificity opens a spectrum of possibilities for non standard forms and building typologies directly related to the specific material used to build them. The close relationship of material and form not only allows for spatial perception but possibly offers models for an optimal structural use of a specific type of wood.

The material doesn't have to fit into a preconfigured structural (standardized) system but instead can be used to its full potential. The assembly of structural components is partially derived from the material geometry and structural performance in a process of form finding that aims at exploring structures that are natural to the material.

The aggregations of irregular elements in a regular topology allows for a first grasping of the essence of branch formations. These also provide the basis for structural analysis models and speculations of usability. Pattern based assemblies are strong structuring devices for spatial organizations. The pattern principles imply an order as a guiding logic for a possible architectural scheme. The dominant role of the pattern in the design process ensures a specific architecture closely related to the logic of the branch formation but the strict order also comes with some limitations.

b) Iterative aggregation / self organisation

In a procedural assembly (Fig. 13) parts are put together in iterative steps. Instead of a global logic controlling the formation local parameters and define the aggregation in every step. There is no fixed or predefined framework that is put together, but the form of the aggregation is found in the process of making.

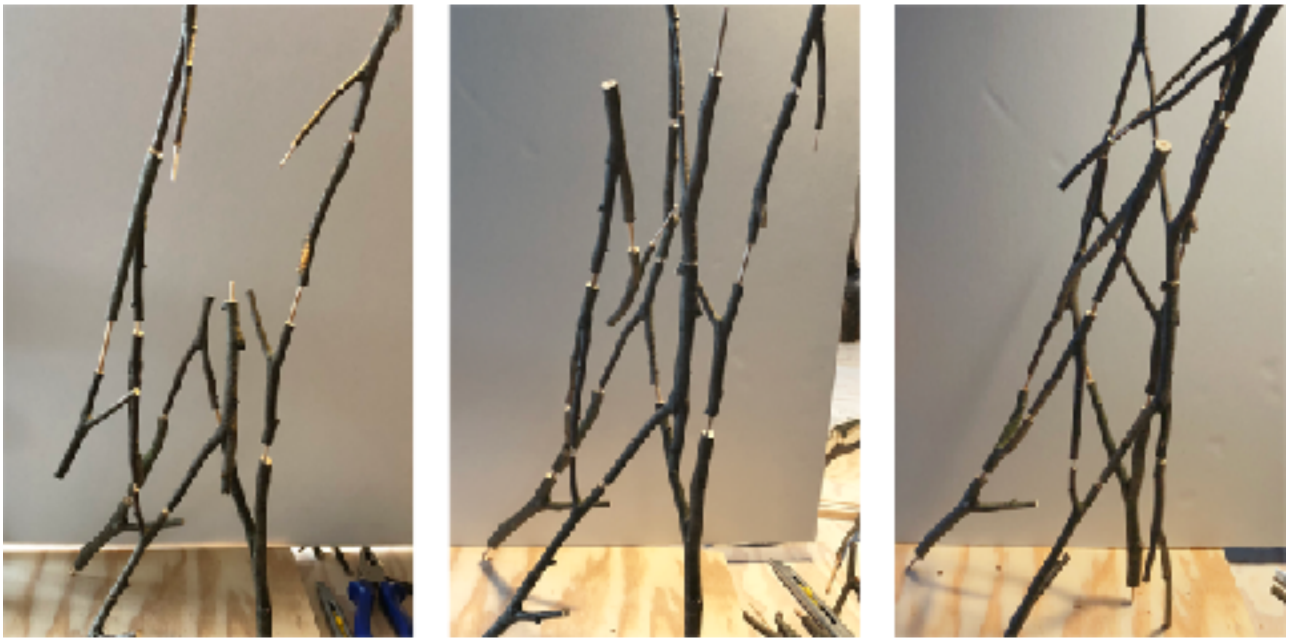


Fig.13: study model: iterative assembly

Self organized structures can be adapted to local conditions by maintaining an overall systematic coherence dependent of the respective ruleset. The resulting irregular structures represent individual solutions to actual problems and conditions as unique formations, a quality very similar to the structure of a tree. Its growth is dependent on the characteristics of its species, but it finds a unique form in relation to its individual growing conditions.

Aggregations based on self organizational principles with their inherent irregularity are more difficult to control and put an additional level of complexity to branch formations. But those offer various possibilities for the addressing of specific problems, whereas the regular topology of principle a) is more restrictive and fixed to its pattern logic. Future investigations in this project will address the development of concepts and algorithms to design and control iterative aggregations.

c) Population of a designed structure:

The nodes of a predefined framework are being replaced by branch elements with crotch or reflex angles matching the respective angles between axes of structural members.
Following principle c) a conventionally designed structural framework composed of generic axes could be modified by replacing structural nodes by forked branches with matching or similar angles. Altered geometrical configurations and the related effect on structural behavior can be analyzed. This concept is currently not in focus.

5. Joining

Considering those formations as physical structures the joining of elements becomes a key problem to address.

In a complex framework typically linear members are connected at the axis nodes. Especially with a high number of members joined, details at the nodes require a high level of precision and technical as well as manufacturing efforts. In such frameworks standardized (simple) members are connected with complex joints.
This project is working with an inverse situation: the non-standard (complex) members represented by branch nodes are connected with simple joints. With the chosen concept of grain continuation all joints are instances of extension joints. The principle of the Y-shaped element offers bifurcating directions which allows for a high level of spatial complexity with only one type of joint.

Structural members are joined at the least complex location within the axis model and only two elements are joined in each location. This way the complexity and specificity of the element is compensated by the geometric simplicity of the joints. The complexity of the non-standard members which is a product of the tree's growth pattern as a representation of the material property of wood reflects in spatial complexity and variation.

5.1. *Joining principles*

To translate the digital aggregates into structures assembled by physical elements we consider two possible concepts:

a) branch elements joined with additional bridging elements:

Adding elements of variable length in between branch nodes allow for flexible aggregations. In each joint two degrees of freedom apply for the combination of elements along the same axis: axial rotation and translation provided by the extension element. Although allowing almost any combination of crotch angles, the insertion of an intermediate bridging element weakens the system structurally by introducing additional joints.

b) branch elements directly joined:

This concept requires a precise selecting and matching of the available specimen. With the limitation of branch lengths also a translation along the connection axis as a degree of freedom is reduced. Although geometric limitations apply, this joining principle provides a maximum stability of a structural system due to minimum number of joints. The goal in this project is to achieve formations assembled with elements joined without extensions.

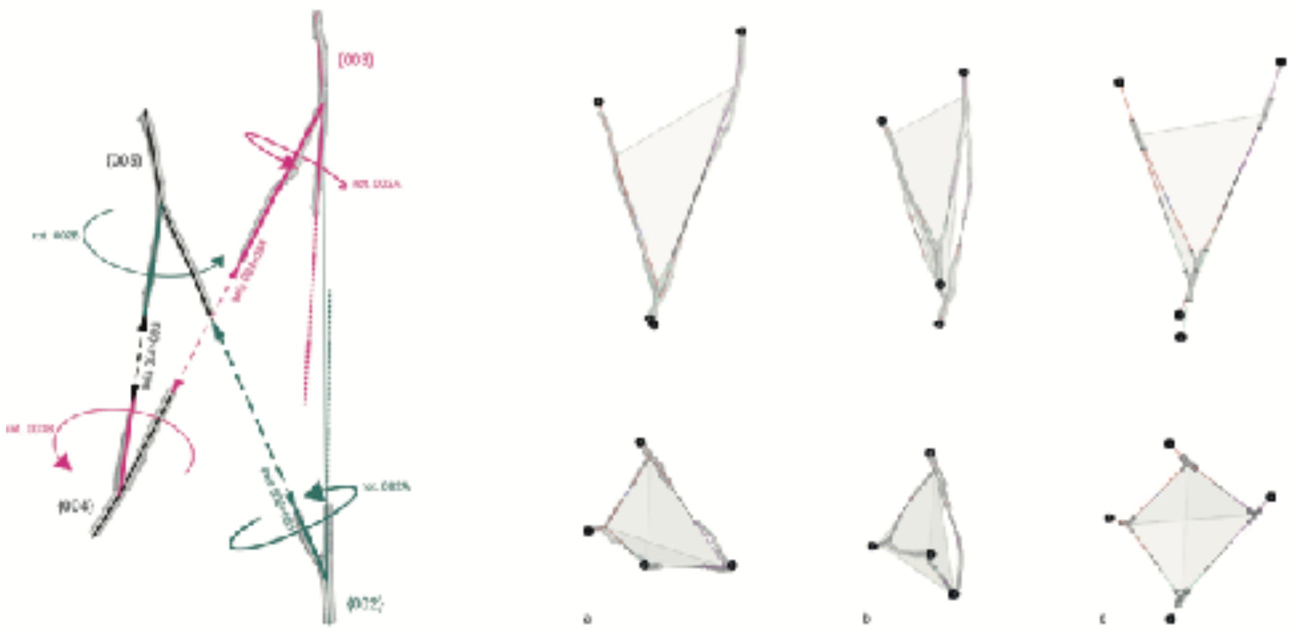


Fig.14: tetrahedron assembly variations in relation to abstraction models (see 3.1.)

5.2. ***Joints***

The joining of branch nodes in the context of this project needs to address the following problems:

Stability:

Within the complex frameworks with differently oriented structural members and loads acting in various directions, the joints need to have the capacity to compensate a set of different forces. Possibly a variation of joints in some instances is needed.

Tolerance:

Given the spatial complexity in the system some tolerance is required to be able to join branch elements into closed cells without applying tension.

Degrees of freedom for assembly:

Combining elements to close a loop requires sufficient degrees of freedom regarding the direction in which parts are put together. Possibly different joint types for “standard” and “closing” joints are needed.

With the intention of developing joints without metal fasteners or glue we chose the Japanese “Kanawa Tsugi” as a basis for a joint study (Fig. 15). This joint is typically used in vertical orientation to extend beams that receive bending loads. It has the benefit of being also relatively stable under application of other loads. This way it is possible to solve all instances with one kind of joint.

Fig.15: joint study

Other joint types and techniques such as other traditional extension joints, steel- and 3D printed joints are studied additionally. Especially current developments in 3D printing technology support speculations on a joining technique that would literally extend wood fibers of the connected elements.
Perhaps concepts can be developed for interpolating grain patterns by 3D printed strands.

6. Structural Evaluation

One of the key interests in Branch Formations is the exploration of the spatial and structural quality inherent in wood as a dynamic material. In this context we are approaching structural design not as a means of solving a given problem but instead we are working with an experimental reversed method: The structural purpose and usability of spatial frameworks are found in an investigative process and developed on the basis of geometric and structural qualities provided by the formations that are defined by principles derived from the grown logic of the material.
Various aggregations of branch node elements are tested with regard to ideal structural performance under influence of different load cases. Potential functions are assigned accordingly and the structural system adapted in a feedback system.

At the current state of the project mainly instances of pattern based aggregations (see also paragraph 4.1.a) are used as structural samples.
A bounding box is populated with a theoretically endless framework of elements and evaluated in a structural simulation with Karamba3d for Grasshopper (Preisinger, 2013). Differently oriented aggregations of the same pattern are tested, each with gravitational loads and supports set at the bottom ends show the maximum deformation in each instance within the same load scenario (Fig. 16). The alternative aggregation samples are evaluated with the goal to find an optimized orientation of the structural members.
The open ends at the borders of the bounding volume appear as the weakest areas. Within the grid, in areas where all elements are interlinked the system shows a much higher stability.

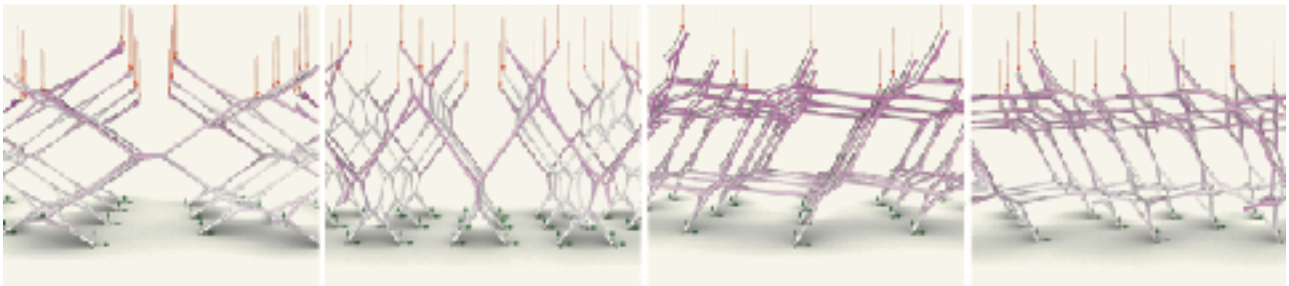


Fig.16: structural analysis: deformation of different orientations of tetrahedron grid

Further studies will investigate to which extent and in which cases irregular systems are structurally superior to regular systems. In a feedback with the development of joints a more in depth analysis of the distributions of forces and moments across the structural model will eventually inform an optimized positioning of joints along the axes in which branch ends are connected. Complementary to a global optimization a structural analysis and simulation could also influence the morphogenesis of the joints themselves. Joining techniques are studied in correspondence with structural behavior and performance.
Besides the analysis of the aggregations as abstract form we additionally conduct an investigation in case studies. With this method the potential of branch aggregations as structural systems are explored in architectural scenarios that relate to a projected usability. The integration of a new geometry is also a first step towards a speculation of scale and use in an architectural sense.

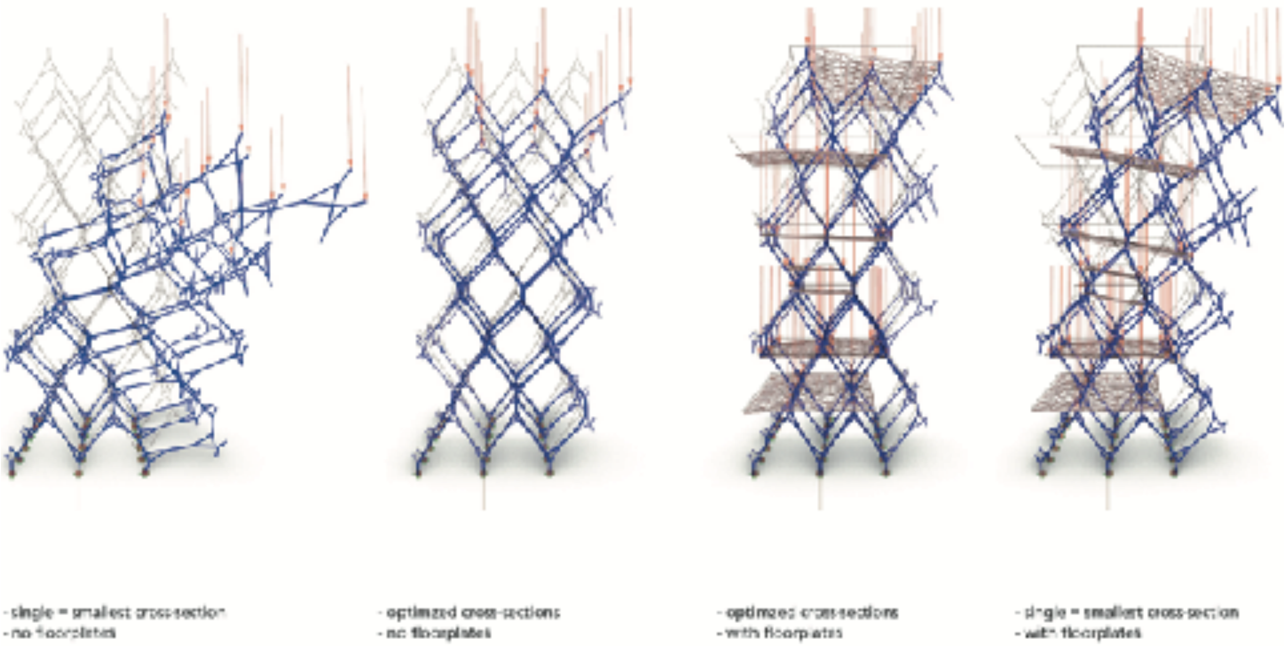


Fig.17: regular tetrahedron-tower load scenarios (gravity and dead weight)

In a first instance the aggregation showing the lowest deformation in the initial testing is used for a case study of a simple vertical architectural scheme (Fig. 17). In this scenario the aggregated branch elements serve as a structure supporting horizontal plateaus. Structural simulation shows that the introduction of floors stabilizes the system and partially solves the problem of open ends at the periphery.

A cross section optimization algorithm distributes a preset range of variable widths for each member across the entire model. This information will be used in a next step to position the specimen from the scanned set with matching cross section according to local structural utilization to form a model with an ideal global structural performance.
The introduction of irregular parts would also transform the regular grid into an irregular model. Different to the previous studies in which a random combination of elements resulted in a transformation of the regular grid (4.1, Fig. 11 and 12) in this study specific local (structural) parameters define the distribution of the varying branch elements and result in an optimized irregularity.

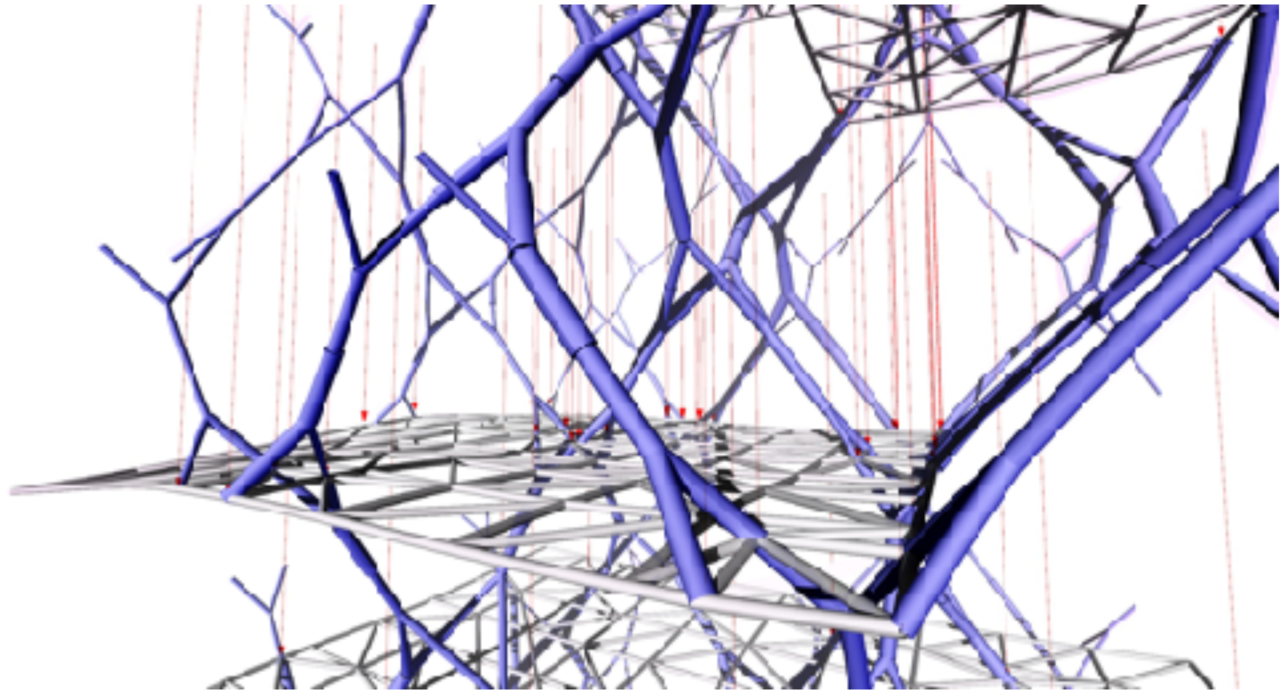


Fig.18: tower case study, close up view

5. Speculation of use

In a similar way as we are working on developing structural models we also intend to find adequate use and architectural function related to the spatial properties and features in the process. Usability and program are not initially set out as a goal to be achieved. With this agenda we start with the exploration of potential of spatial configurations and develop a possible functionality in a process of negotiating geometry and use. We aim at finding out more about the relationship between architecture and the material wood but also at finding novel modes of experience, usability and architectural typology, a new materialist wood architecture across multiple scales from furniture to high-rise building.

6. Conclusion and Outlook

A key quality and challenge of the proposed method is the tension between control and coincidence.
Further research will investigate structural performance and potential function of various aggregations as well as methods of structural optimization.
With the aim of arriving at full scale prototypes there is an ongoing development of joint details for optimal load transmission.

The researchers will further explore the relationship of the angle spectrums occurring in various tree species and geometric and structural possibilities and limitations.

“Branch Formations” represents the exploitation of a distinct conceptual agenda, but could be seen as one case study in the broader context of a new understanding of architectural and structural design using wood in its naturally grown form. With constantly improving computation and technology paired with a growing awareness for renewable resources wood doesn’t need to be processed into a homogeneous material but can be addressed and utilized in its full complexity.

REFERENCES

1. U. Müller, W. Gindl-Altmutter, J. Konnerth, D. Kaserer, J. Keckes (2014) Stamm-Astanbindung – eine biologisch optimierte Struktur mit hoher mechanischer Leistungsfähigkeit. 10.GraHFT'14
2. Shigeru Ban Architects (2010) Centre Pompidou, Metz, France
3. M. Self, E. Vercruysse (2016) Wood Chip Barn. Design + Make. The Architectural Association London
4. K. Spindler (1995) cf wooden handle of the cooper axe used by the neolithic hunter Otzi 5.000 years ago. In: Der Mann aus dem Eis, Goldmann
5. X. Agote Aiziprua (2009) ct timber suppliers for shipyards. In: Bertain#23 'from tree to ship' p109 Departamento de Cultura y Euskera, 2009
6. C. Schindler, M. Tamke, A. Tabatabai, M. Bereuter (2013) Serial Branches. In: Proceedings of eCAADe 31-Computation and Performance-Volume1-Material Studies, 2013.
7. S. Schüler, M. Grabner, S. Karanitsch-Ackerl, S. Fluch, R. Jandl, T. Geburek, H. Konrad (2013) Fichte - fit für den Klimawandel? BFW-Praxisinformation 31: p10 - p12
8. M. Hanewinkel (2010) Baumarteneignung Fichte und Buche bei Klimawandel. http://www.waldwissen.net
9. G. Lüster (1913) Das Rheingauer Gebück. In: Naturdenkmäler in Nassau 2. Bechtold, Wiesbaden, 1913
10. M. Grabner (2017) WerkHolz. Kessel, Remangen-Oberwinter
11. A. Rossi (2017) [www.food4rhino.com/app/wasp]
12. D. Piker (2017) [www.food4rhino.com/app/kangaroo-physics]
13. C. Preisinger (2013) Linking Structure and parametric Geometry. In: Architectural Design, 83: p110-113, doi: 10.1002/ad.1564, 2013.

Acknowledgements

This work is part of the PEEK research project Conceptual Joining (AR 395-G24) funded by the Austrian Science Fund (FWF). Christoph Kaltenbrunner and our fellow team members contributed with expertise and feedback. Philipp Reinsberg participated with joint studies. Matthew Tam of Bollinger + Grohmann engineers contributed with an initial Karamba definition. Sandra Tuider provided the branch samples. The Dept. Of Geometry of the Angewandte made available an Artec EVA 3D Scanner.